중앙대학교

NightAdapter: Learning a Frequency Adapter for Generalizable Night-time Scene Segmentation

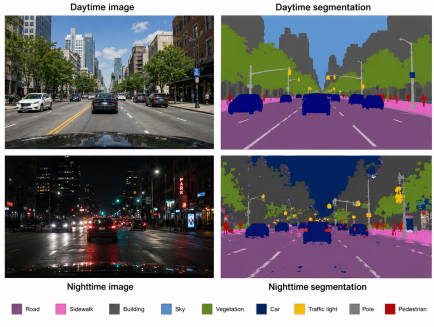
박상춘

응용통계학과 중앙대학교

2026년 05월 29일

연구 배경: Night-time Scene Segmentation

- Semantic segmentation은 자율주행 scene understanding의 핵심 task
- Day-time 환경에서는 segmentation 성능이 크게 발전했지만, night-time 환경에서는 성능 저하가 발생
- 주요 원인: **복잡한 조명 환경**
  - low illumination, glare, shadow
  - low contrast, over/under-exposure
- 따라서 night-time segmentation에서는 **illumination robustness**가 중요



day-time과 night-time segmentation 차이 예시

왜 Night-time Generalization이 어려운가?

- ''Night''는 하나의 고정된 domain이 아니다.
- 서로 다른 night datasets는 모두 night-time scenes이지만, illumination style이 크게 다름

Night Domain A	**Night Domain B**	**Night Domain C**
가로등이 많고 밝은 장면	헤드라이트와 glare가 강한 장면	매우 어둡고 contrast가 낮은 장면

A에 맞춰진 모델이 B, C에서도 잘 동작한다고 보장하기 어렵다.

기존 Paradigm과 한계

Fully-Supervised Paradigm

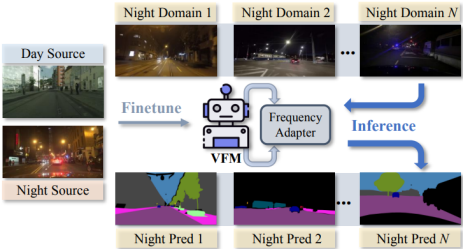
- 라벨 된 night images로 segmentation model 학습
- 같은 night domain에서는 강할 수 있음
- 그러나 train/test night scenes가 i.i.d.라는 가정에 의존

Domain Adaptation Paradigm

- Day-time source 정보를 target night domain으로 transfer
- Target night domain에 맞춰 성능을 개선할 수 있음
- 그러나 Training phase에서 target domain이 관측 가능하다는 가정에 의존

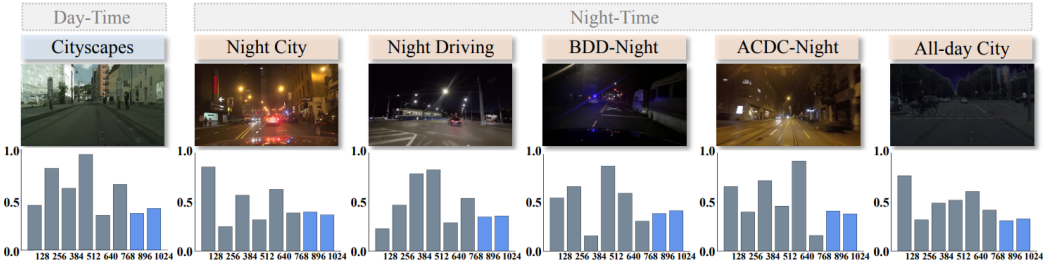
핵심 한계: 학습 때 보지 못한 illumination conditions에 대한 generalization이 부족하다는 것

Problem Definition and Motivation



- Source domain은 day-time 또는 night-time scene
- Inference에서는 학습 때 보지 못한 night-time domains가 입력
- 입력은 **domain-variant**하지만, 예측은 **domain-agnostic**이어야 함
- **목표**: 학습 때 보지 못한 다양한 night-time scene에도 일반화되는 segmentation model

Frequency-Band Observation



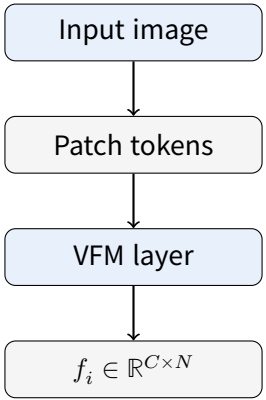
- VFM feature를 frequency domain에서 보면 band별로 day/night pattern이 다르게 나타남
- **Low-frequency bands**: day-time과 night-time에서 상대적으로 유사한 pattern
- **Middle/high-frequency bands**: domain-specific illumination variation이 큼
- 이 관찰이 band별로 다르게 처리하는 frequency adapter의 동기

Input to NightAdapter: VFM Feature

- NightAdapter는 raw RGB image가 아니라 **VFM layer feature**에 작동
- Backbone은 Vision Foundation Model인 **DINOv2**를 사용
- i-th VFM layer feature:

$$f_i \in \mathbb{R}^{C \times N}$$

- C: feature channel dimension
- N: number of patch tokens



Discrete Sine Transform and Band Split

- Frozen VFM feature를 DST로 frequency domain에 투영

$$f_i \xrightarrow{\text{DST}} V_i, \quad V_i \in \mathbb{R}^{C \times N}$$

- 마지막 DINOv2 encoder feature에서 $C = 1024, N = 1024$
- C 방향을 128 단위로 나누어 8개 frequency band를 만듦

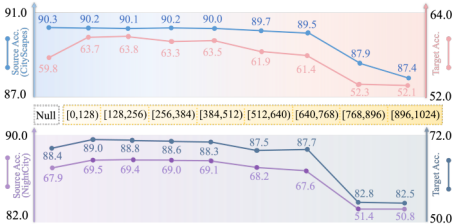
Band range	Interpretation
$[0, 256)$	High-frequency (2)
$[256, 768)$	Middle-frequency (4)
$[768, 1024)$	Low-frequency (2)

$$V_i(u,v) = \sum_{c=0}^{C-1} \sum_{n=0}^{N-1} f_i(c,n)\, \alpha_u(c)\, \alpha_v(n)$$

$$\alpha_u(c) = \sin\left[\frac{\pi(u+1)(c+1)}{C+1}\right]$$

$$\alpha_v(n) = \sin\left[\frac{\pi(v+1)(n+1)}{N+1}\right]$$

Band Rejection Analysis



- 특정 frequency band를 하나씩 제거하며 source/target accuracy 변화를 분석
- High/middle-frequency band 제거 시 unseen target accuracy가 개선되는 경향
- Low-frequency band 제거 시 source와 target accuracy가 모두 하락
- Low band는 공통 scene content를 담고, high/middle band는 illumination 변화에 더 민감한 것으로 해석

Divide-and-Conquer Strategy

Low-frequency bands

- Illumination-insensitive scene content / layout

⇒ **Illumination-Insensitive Band Adaptation (IBA)**

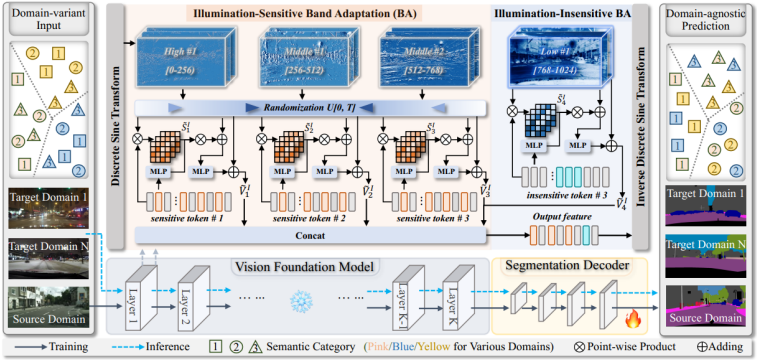
Middle/high-frequency bands

- Illumination-sensitive style / edge / detail

⇒ **Illumination-Sensitive Band Randomization (SBR) + Illumination-Sensitive Band Adaptation (SBA)**

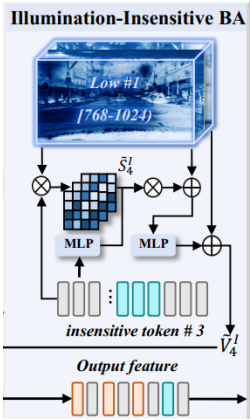
- **서로 다른 band는 서로 다른 방식으로 adaptation 되어야 함**

Overall Architecture



- VFM layer에 삽입되는 frequency adapter로 구현
- Adaptation된 band들은 IDST로 다시 복원되어 다음으로 전달

IBA: Illumination-Insensitive Band Adaptation



Low-frequency band를 illumination-insensitive 방향으로 adaptation

$$V_i^L = [V_i^{[768,896)},\ V_i^{[896,1024)}]$$

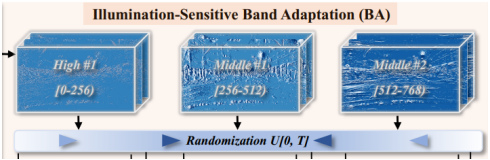
$$S_i^L = \mathrm{Softmax}\left(\frac{V_i^L (T_i^L)^T}{\sqrt{C}}\right)$$

$$\tilde{f}_i^L = S_i^L [T_i^L W_i^1 + b_i^1]$$

$$\tilde{V}_i^L = V_i^L + (\tilde{f}_i^L + V_i^L) W_i^2 + b_i^2$$

- T_i^L: learnable token으로 content prior 추출
- S_i^L: low-frequency patch와 token 사이의 similarity map
- $\tilde{f}_i^L$: token이 담은 content prior를 feature space로 projection
- $\tilde{V}_i^L$: 기존 low-frequency band와 projected token feature를 fusion

SBR: Illumination-Sensitive Band Randomization



- High/middle band는 illumination-sensitive 하여 domain gap을 키움
- 단순 제거 시 edge/detail 정보까지 손실될 수 있음
- SBR은 response를 제한된 범위에서 randomize

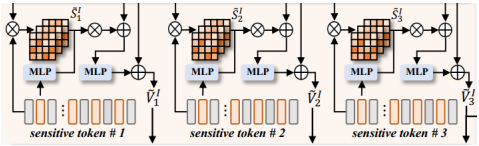
Band grouping

$$V_i^H = [V_i^{[0,128)}, V_i^{[128,256)}]$$
$$V_i^{M1} = [V_i^{[256,384)}, V_i^{[384,512)}]$$
$$V_i^{M2} = [V_i^{[512,640)}, V_i^{[640,768)}]$$

Per-channel randomization

$$V_i^{j,k} \sim U[0,T], \quad j = 0, \dots, 767,\ k = 0, \dots, 1023,\ T = 0.3$$

SBA: Illumination-Sensitive Band Adaptation



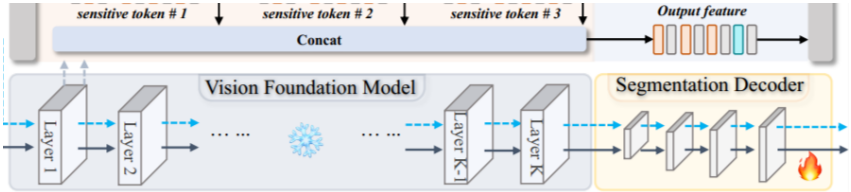
- SBR로 randomize된 sensitive band를 다시 adaptation
- Band별 learnable sensitive token 사용
  $T_i^H,\ T_i^{M1},\ T_i^{M2}$
- Edge/detail은 유지하고 illumination-specific 변동은 감소

High-frequency group 예시

$$S_i^H = \mathrm{Softmax}\left(\frac{V_i^H (T_i^H)^T}{\sqrt{C}}\right)$$
$$\tilde{f}_i^H = S_i^H [T_i^H W_i^3 + b_i^3]$$
$$\tilde{V}_i^H = V_i^H + (\tilde{f}_i^H + V_i^H) W_i^4 + b_i^4$$

Middle groups도 같은 방식으로 $\tilde{V}_i^{M1}$, $\tilde{V}_i^{M2}$ 생성

Band 복원과 예측



- IBA, SBR, SBA가 끝난 band들을 다시 concatenate
- IDST를 통해 spatial feature로 복원
- 복원된 feature는 다음 frozen VFM layer 또는 segmentation decoder로 전달

Experimental Setup

구현 설정

- Encoder: DINOv2
- Decoder: Mask2Former decoder
- Input size: 512×512
- Optimizer: AdamW
- Training: 40K iterations
- Metric: mIoU

평가 설정

1. Day-to-night generalization
2. In-domain night-time segmentation
3. Night-to-night generalization

데이터셋

- Cityscapes, NightCity, NightCity-fine
- ACDC-Night, Dark Zurich, Nighttime Driving
- BDD100K-Night, All-day CityScapes

Day-to-Night Generalization

Method	Backbone	AN	DZ	ND
DA Models:				
GCMA [60] [ICCV'2019]	Res-101	-	42.0	45.6
MCGDA [61] [PAMI'2019]	Res-101	-	42.5	49.4
DANNet [72] [CVPR'2021]	Res-101	-	44.3	47.7
CCDistill [20] [CVPR'2022]	Res-101	37.3	47.5	-
UDA Models:				
AdaptSeg [68] [CVPR'2018]	Res-101	-	30.4	34.5
TENT [69] [ICLR'2021]	Res-101	25.2	21.6	39.3
DAFormer [28] [CVPR'2022]	MiT-B5	44.7	48.5	51.8
MALL-D [58] [ECCV'2022]	Res-101	26.9	22.8	42.7
SePiCo [73] [PAMI'2023]	Res-101	50.5	54.2	56.9
HRDA [29] [ECCV'2022]	MiT-B5	53.1	55.9	-
STA [23] [MM'2023]	MiT-B5	49.7	-	-
MIC [30] [CVPR'2023]	MiT-B5	57.2	60.2	58.6
PASS [43] [AAAI'2024]	MiT-B5	60.3	60.2	57.0
CoDA [24] [ECCV'2024]	MiT-B5	66.4	61.2	59.2
DGInStyle [38] [ECCV'2024]	MiT-B5	59.2	32.6	-
REIN [71] [CVPR'2024]	ViT-L	70.6	64.3	50.3
NightAdapter (Ours)	ViT-L	**84.0**	**74.1**	**60.3**
		↑13.4	↑9.8	↑1.1

- Train: Cityscapes day-time source
- Test: ACDC-Night, Dark Zurich, Nighttime Driving
- 모든 DA/UDA 방법들보다 NightAdapter는 더 높은 성능
- REIN 대비 AN, DZ, ND 모두에서 큰 mIoU 향상

In-domain Night-time Segmentation

Method	#Para.	NC	NC-f	BN
ResNet-101 Encoder:				
DeepLabV3 [10] [ECCV'2018]	60.1M	54.7	58.8	43.4
NightCity [65] [TIP'2021]	84.6M	51.8	55.9	39.7
NightLab [17] [CVPR'2022]	98.5M	55.9	62.3	50.4
DTP [70] [ICCV'2023]	63.9M	57.6	60.4	-
NightFormer [55] [ECCV'2024]	58.4M	59.8	62.8	47.5
Swin-Base Encoder:				
UPer-Swin [49] [ICCV'2021]	119.9M	58.4	61.1	31.7
NightLab [17] [CVPR'2022]	242.4M	59.8	62.3	45.1
FDLNet [75] [TIP'2023]	-	55.5	-	45.9
DTP [70] [ICCV'2023]	122.5M	61.2	64.2	53.1
NightFormer [55] [ECCV'2024]	111.9M	63.5	65.9	55.4
VFM Encoder:				
REIN [71] [CVPR'2024]	3.0M	62.4	64.6	53.9
NightAdapter (Ours)	11.5M	**65.1**	**66.2**	**57.8**
		↑1.6	↑0.3	↑2.4

- Train/Test를 같은 night-time dataset에서 수행
- Datasets: NightCity, NightCity-fine, BDD100K-Night
- DTP, NightFormer와 같은 night-specific method와 비교
- In-domain 설정에서도 경쟁력 있는 성능을 보임

Night-to-Night Generalization

Method	Backbone	NC as Source					NC-f as Source				
		AN	DZ	ND	BN	AC	AN	DZ	ND	BN	AC
RobustNet [14] [CVPR'2021]	Res-50	25.3	22.9	36.4	21.6	23.9	28.9	25.3	38.6	22.4	27.1
SHADE [83] [ECCV'2022]	Res-50	27.0	23.4	37.8	22.5	26.3	29.6	27.7	39.4	23.6	27.7
SAW [57] [CVPR'2022]	Res-50	26.8	24.1	38.2	23.0	27.2	30.3	27.9	39.9	24.5	29.0
WildNet [41] [CVPR'2022]	Res-50	26.1	23.5	37.4	22.6	25.8	29.1	26.4	38.2	23.9	28.3
SPC [33] [CVPR'2023]	Res-50	28.6	26.4	38.9	25.0	28.2	32.5	30.3	41.7	28.8	31.9
HGFormer [19] [CVPR'2023]	Swin-B	33.7	32.2	44.4	32.8	33.7	38.8	35.6	50.8	36.9	35.4
CMFormer [6] [AAAI'2024]	Swin-B	35.2	33.9	45.7	33.5	34.0	40.6	37.9	50.4	37.5	38.2
BlindNet [1] [CVPR'2024]	Res-50	27.4	25.0	37.5	23.8	24.6	30.3	28.7	39.1	26.8	28.6
SET [79] [MM'2024]	ViT-L	45.3	44.0	61.2	33.6	42.9	48.8	47.1	62.5	38.9	47.3
DGInStyle [38] [ECCV'2024]	MiT-B5	40.7	37.6	48.4	32.9	39.5	43.1	40.4	54.3	35.7	41.8
tqdm [54] [ECCV'2024]	ViT-B	42.5	41.8	60.7	31.0	40.7	46.4	44.5	58.6	36.3	44.5
FADA [4] [NeurIPS'2024]	ViT-L	42.9	41.3	60.9	34.0	43.2	49.1	45.6	62.0	39.1	46.4
REIN [71] [CVPR'2024]	ViT-L	43.6	42.4	59.7	34.8	41.1	50.7	46.2	60.8	37.5	45.8
NightAdapter (Ours)	ViT-L	**50.5**	**47.0**	**62.3**	**40.5**	**46.9**	**54.8**	**50.1**	**64.0**	**42.3**	**50.4**
		↑5.2	↑3.0	↑1.1	↑5.7	↑3.7	↑4.1	↑3.0	↑1.5	↑3.2	↑3.1

- Source는 NightCity 또는 NightCity-fine, target은 unseen night domain
- NightAdapter는 최신 DGSS 및 VFM 기반 방법들보다 일관되게 높은 성능을 보임

Ablation Study

Components			Trained on Cityscapes		
IBA	SBR	SBA	AN	DZ	ND
✗	✗	✗	70.6	64.3	50.3
✓	✗	✗	72.5	67.8	52.7
✓	✓	✗	76.0	70.4	54.8
✓	✗	✓	74.9	69.1	53.4
✓	✓	✓	**84.0**	**74.1**	**60.3**

Table 5. Ablation studies on each component.

Frequency Bands				Trained on CitySCapes		
$\mathcal{V}_i^L$	$\mathcal{V}_i^{M_1}$	$\mathcal{V}_i^{M_2}$	$\mathcal{V}_i^H$	AN	DZ	ND
✗	✗	✗	✗	70.6	64.3	50.3
✓	✗	✗	✗	72.5	67.8	52.7
✓	✓	✗	✗	74.8	69.7	54.6
✓	✓	✓	✗	77.2	71.5	56.7
✓	✓	✓	✓	**84.0**	**74.1**	**60.3**

Table 6. Ablation studies on each frequency band.

Component ablation

- IBA만 사용해도 REIN 대비 성능이 향상
- SBR과 SBA를 추가하면 성능이 더 향상
- 모든 component를 함께 사용한 full model이 가장 좋은 결과

Frequency band ablation

- Low-frequency adaptation은 유용하지만 충분하지 않음
- Middle/high-frequency band도 함께 adaptation해야 함

Conclusion

- **VFM feature의 frequency-band 분석**
  - DST를 통해 illumination-sensitive band와 illumination-insensitive band를 구분
- **Band-wise frequency adapter**
  - IBA는 low-frequency illumination-insensitive band를 강화
  - SBR + SBA는 middle/high-frequency illumination-sensitive band를 randomize하고 adaptation 진행
- **강한 generalization 성능**
  - Day-to-night, night-to-night, in-domain setting에서 기존 방법보다 높은 성능을 보임
- **Night-time generalization은 illumination sensitivity에 따라 frequency band를 다르게 처리함으로써 개선될 수 있다.**